

Clarification Response – IEEE Tech4Good 2025

Project Title: IoT-based Hydroponic Farming and Solar Energy Integration for Sustainable Agriculture

Proposal ID: 25-ComT4G-51

Submitted by:

Hilal Hudan Nuha PhD. & Hassan Rizky Putra Saillellah

Telkom University, Indonesia

Date: October 2025

1. Implementing Team Technical Qualifications and Relevant Experience

The implementing team comprises experienced lecturers and researchers from Telkom University, with strong expertise in IoT systems, renewable energy, automation, and community empowerment. Each team member has relevant project experience, academic publications, and direct field engagement in IoT-based agriculture.

- **Hilal Hudan Nuha PhD.**

IEEE Senior Member (No. 96162702)

Role: Project Lead – IoT and Renewable Energy System Integration

Affiliation: Telkom University – Faculty of Informatics

Technical Expertise:

IoT systems for environmental monitoring, renewable energy integration, embedded systems, and sustainable automation.

Relevant Experience:

- *Implementation of monitoring and prediction of humidity, temperature, and light using Gaussian Process Regression (GPR) for orchid greenhouses in Lembang.* (Published work – Google Scholar)
- *Automation of IoT-based COVID-19 Isolation Room Temperature and Humidity Control System* at Telkom University. (Published work – Google Scholar)
- *Microhydro Power Plants with IoT-based Monitoring and Prediction using Linear Regression.* (Published work – Google Scholar)
- Ongoing **patent under verification**: *Indoor Hydroponic Farming System using Growlamp Lighting for Urban Agriculture* (solar-light replacement innovation).
- Has extensive experience in deploying IoT systems for environmental and agricultural use cases, integrating hardware, sensors, and predictive algorithms for sustainable automation.

- **Hassan Rizky Putra Sailallah**

IEEE Member (No. 101270452)

Role: IoT System Development Lead

Affiliation: Telkom University – IoT Laboratory Coordinator – Faculty of Informatics

Technical Expertise:

IoT network integration, precision agriculture systems, real-time monitoring and automation, microcontroller-based system development.

Relevant Experience:

- *Lead for AyoBerAKSI 2025* project titled “**Smart Hydroponic Solutions: IoT Integration for Efficient Urban Agriculture.**”
- Coordinator of **Telkom University Internet of Things Laboratory**, focusing on IoT-based agricultural innovations.
- Hands-on hydroponic practitioner with direct experience in nutrient solution management, pH/ppm calibration, and greenhouse automation workflows.
- Has guided multiple student projects in hydroponic system design and sensor calibration for sustainable farming.

- **Muhammad Irsan PhD.**

IEEE Member (No. 100311278)

Role: Sensor and Actuator Integration Engineer

Affiliation: Telkom University – Faculty of Informatics

Technical Expertise:

IoT sensor design, embedded system integration, cloud computing for agriculture, automation control.

Relevant Experience:

- *Supervisor for Innovillage 2024* project: “**Optimization of Hydroponic Cultivation using Cloud Computing and IoT with Greenhouse Implementation at GR Farm.**”
 - Lessons learned from structural challenges (greenhouse collapse due to heavy rain) and system usability feedback (mobile-first redesign for 2025 iteration).
- *Lead for DRTPM Grant: Driver Drowsiness Detection and Post-Accident Integrated System.*
- *PhD Dissertation: A Novel Prototype for Safe Driving using Embedded Smart Box System (IoT-based Safety Monitoring).*
- *Optimized Attendance System using IoT-based RFID for SDN 2 Midang and IoT-based Chrome Tank Water Temperature Control System.* (Published work – Google Scholar)
- Strong experience in embedded system deployment and real-world data integration.

• **Muhammad Faris Fathoni PhD.**

IEEE Member (No. 99226510)

Role: UI/UX and Software Development Lead

Affiliation: Telkom University – Embedded System Laboratory

Technical Expertise:

Mobile and web app development, machine learning in agriculture, embedded software testing, and predictive analytics.

Relevant Experience:

- *Supervisor for Innovillage 2024 project on **Automatic Watering System for Strawberry Plants in Lembang.***
- *Rice Plant Disease Detection using Transfer Learning with MobileNetV3-Large. (Published work – Google Scholar)*
- *Advisor for Embedded Hardware Laboratory focusing on real-time IoT control and automation systems. (Published work – Google Scholar)*
- *Forecasting Fertilizer Level using PSO and LSTM for Nutrient Efficiency. (Published work – Google Scholar)*
- Experience in designing real-time dashboards and mobile applications for IoT systems, ensuring accessibility for farmers with low digital literacy.

• **Dr. Aji Gautama Putrada**

IEEE Member (No. 93865067)

Role: Renewable Energy Engineer & Solar Power Integration Specialist

Affiliation: Telkom University – Faculty of Informatics

Technical Expertise:

Solar energy systems, smart lighting control, sensor fusion, and edge computing for IoT-based automation.

Relevant Experience:

- *Machine Learning Methods in Smart Lighting Towards Achieving User Comfort: A Survey. (Published work – Google Scholar)*
- *CIMA: A Novel Classification-Integrated Moving Average Model for Smart Lighting Control. (Published work – Google Scholar)*
- *Optimization of Grow Light Control in IoT-Based Aeroponic Systems with Sensor Fusion and Random Forest Classification. (Published work – Google Scholar)*
- *Predictive Control on Lettuce NFT-Based Hydroponic IoT Using Deep Neural Network. (Published work – Google Scholar)*
- *Evaluation of IoT-based Grow Light Automation on Hydroponic Plant Growth. (Published work – Google Scholar)*
- *Q8KNN: A Novel 8-Bit KNN Quantization Method for Edge Computing in Smart Lighting Systems with NodeMCU. (Published work – Google Scholar)*
- Extensive background in renewable energy integration and energy-efficient IoT design.

- **Student Members (Telkom University – Faculty of Informatics)**

- Junior Developers and Field Engineers**

- Role:** IoT hardware development, software engineering, mobile and backend programming.

- Affiliation:** Telkom University – Faculty of Informatics

- Team:**

- **2 students** – *Mobile app development for real-time hydroponic monitoring and control.*
 - **2 students** – *Backend system development (database, API, dashboard connectivity).*
 - **6 students** – *IoT hardware assembly and testing (pH, TDS sensors, microcontroller integration, relay and pump automation).*
 - *All student members receive technical mentoring and are trained to document and maintain the deployed systems as part of their applied learning experience.*

- **GR Farm Team (Community Partner)**

- Hydroponic Operations and Community Trainers**

- Role:** Agricultural operations, crop maintenance, and local training facilitation.

- Affiliation:** GR Farm – Yayasan Gowa Raya

- Team:**

- **8 GR Farm members** *will be trained in IoT device operation, solar system maintenance, and community-led hydroponic education.*
 - *They will assist in user training sessions for local residents and ensure daily monitoring of the hydroponic production process after project deployment.*

2. Measurable Outcomes and Income Generation

1. Projected Crop Yield and Revenue Generation

The existing hydroponic system yields approximately **240–270 kg of vegetables per table per year**, at a market price of **USD 2–3 per kilogram**.

With the current setup of 8 tables, the annual production reaches **1,920–2,160 kg**, generating an income of **USD 5,760–6,480** per year.

After expanding to **28 tables**, the total annual yield is expected to reach **6,720–7,560 kg**, generating between **USD 20,160 and USD 22,680** annually.

The integration of **solar panels** significantly reduces electricity expenses—currently around **USD 35–70 per month**—saving approximately **USD 420–840 per year**.

These savings will be redirected to maintenance, sensor calibration, and future system expansion.

2. Revenue Utilization and Reinvestment Plan

To ensure financial sustainability, all revenue from vegetable sales and cost savings will be reinvested based on the following allocation:

Source of Income / Savings	Allocation Purpose	Percentage
Vegetable sales (main revenue)	Reinvestment in system maintenance, sensor replacement, and hydroponic expansion	70 %
Vegetable sales	Incentive distribution to local community members involved in seeding and farming activities	20 %
Vegetable sales	Reserve fund for equipment replacement and emergency maintenance	5 %
Solar power savings	Added to the operational and maintenance budget	5 %

This reinvestment model ensures that maintenance, upgrades, and future community training can be fully funded internally—without the need for additional external grants.

3. Measurable Impact Indicators

Indicator	Baseline (Current)	Target (After 12 Months)	Verification Method
Number of operational hydroponic tables	8 tables	28 tables	Physical verification and photographic documentation
Annual vegetable yield	1,920–2,160 kg	6,720–7,560 kg	Harvest records
Annual income	USD 5,760–6,480	USD 20,160–22,680	Sales receipts and bookkeeping
Annual energy savings	0	USD 420–840	Electricity bill comparison
Number of trained farmers and community members	0	≥ 10 individuals	Attendance sheets
Female participation in training	—	≥ 40 %	Gender-disaggregated attendance records

Indicator	Baseline (Current)	Target (After 12 Months)	Verification Method
Community income from shared revenue	—	~ USD 4,000 annually	Financial distribution records
Operational continuity	—	≥ 3 years sustained operation	Maintenance logs and site reports
Participant satisfaction index	—	≥ 80 % positive feedback	Post-training surveys

These metrics provide quantifiable evidence of progress in productivity, energy efficiency, gender inclusion, and local economic growth.

4. Community and Household Impact (Bonto-Bontoa Village)

Based on demographic data from the **Somba Opu District Statistical Report (2024)**, the gender ratio in Bonto-Bontoa is approximately **50 % male and 50 % female**, similar to the broader Gowa Regency profile. The village is estimated to include **around 800–1,000 households**, primarily engaged in small-scale agriculture and local trade.

By expanding hydroponic capacity and establishing community seedling cooperatives, the project aims to directly benefit at least **10 households in the first phase**, with indirect benefits to **50–60 households** through seedling sales, training programs, and access to lower-cost vegetables.

5. Long-Term Economic Impact

- The combined annual revenue and cost savings (**USD 20,580–23,520**) will fully cover all operational expenses, ensuring **self-sufficiency**.
- The model establishes a **sustainable income cycle**, in which hydroponic revenue supports IoT system calibration, community training, and local employment.
- Female participants in seedling production will gain **consistent micro-entrepreneurship income**, contributing to household financial resilience.

3. Community Technical Capacity and Sustainability

1. Community Technical Capacity Development

To ensure that the IoT-based hydroponic and solar-powered farming system remains functional and sustainable after the IEEE Tech4Good funding period, the project includes a **structured capacity-building program** for both technical and community partners:

Training Focus	Target Participants	Description and Objectives
IoT Device Operation and Maintenance	4 GR Farm operators and 10 local residents from Yayasan Gowa Raya	Training on sensor calibration (pH, TDS), device troubleshooting, and mobile app use. Participants will learn to reset thresholds, read error codes, and conduct regular maintenance checks.
Solar Energy Maintenance and Safety	4 technical staff from Yayasan Gowa Raya	Practical workshop on inverter and battery care, safety inspection, and energy efficiency monitoring. Delivered with supervision from the Faculty of Informatics, Telkom University.
Data Monitoring and Decision Support	Local youth and student volunteers	Training on data interpretation for pH and nutrient balance management using the mobile dashboard, enabling informed farming decisions.
Train-the-Trainer Model	2 GR Farm leaders + 1 Yayasan Gowa Raya technician	Participants are trained to become local trainers to continue technical knowledge transfer beyond the project period.

By the end of the 6-month implementation period, **at least 12 trained individuals** will be certified by the project team as *Local Technical Maintainers* for IoT hydroponic and solar systems.

They will receive both documentation manuals (in Indonesian) and digital tutorial modules hosted on Telkom University's open-access learning platform.

2. Long-Term Maintenance Model

After the grant period, the sustainability structure is designed to be **self-sufficient**, as follows:

Component	Responsible Party	Maintenance Mechanism
IoT System (sensors, controller, mobile app)	GR Farm + Yayasan Gowa Raya	Scheduled calibration every 3 months; backups stored on Telkom University's cloud for redundancy.
Solar Energy System (panels, inverter, battery)	Yayasan Gowa Raya	Routine cleaning and battery cycle checks every 6 months.

Component	Responsible Party	Maintenance Mechanism
Software & Cloud Backend	Faculty of Informatics, Telkom University	Continued hosting on the university's academic server; updates integrated by student volunteers and research assistants.
Training and community expansion	Yayasan Gowa Raya & GR Farm	New local trainees funded by seedling revenue and vegetable sales; 10% of annual net income allocated to the training fund.

This arrangement ensures that **each stakeholder has a clear operational role**, reducing dependency on external technical support.

3. Financial Sustainability from Farm Revenue

The farm's income and operational savings (as outlined in the measurable outcomes section) directly fund long-term system upkeep:

- **5% of total annual revenue** (~USD 1,000 per year) will be specifically reserved for **IoT sensor calibration and component replacement** (e.g., pH electrodes, TDS probes, relay modules).
- **5% of total annual revenue** will fund the **community training program**, enabling Yayasan Gowa Raya to recruit new members and conduct local workshops.
- The **70% reinvestment allocation** already planned in the revenue model (see measurable outcomes) covers broader system maintenance, including pump or panel replacement.

All maintenance costs are therefore internalized within the farm's financial cycle, ensuring **zero dependency on future external grants**.

4. Institutional Support and Technical Oversight

To guarantee continuous improvement and technical reliability:

- The **Faculty of Informatics, Telkom University** will remain a long-term partner, providing **yearly inspection visits** and **remote assistance** through the IoT monitoring platform.
- The system's **open-source nature** allows the community to replace or upgrade sensors using affordable local components.
- Future student projects under Telkom University's "Community-Based IoT" program will integrate into this site as part of their academic practicum, ensuring ongoing innovation and research continuity.

This academic partnership embeds the project within a **sustainable educational ecosystem**, bridging university research with community empowerment.

5. Summary of Sustainability Strategy

Aspect	Mechanism	Duration of Sustainability
Technical skills	Local technician training (train-the-trainer model)	≥ 3 years
Financial funding	10% of annual income allocated to O&M	Continuous
Hardware maintenance	Periodic calibration + Telkom University oversight	Continuous
Solar energy	Zero operating cost after setup	≥ 10 years
Knowledge transfer	Yearly student involvement + digital module	Ongoing

4. Revenue Collection, Maintenance, and Upgrade Governance

1. Governance and Management Structure

To ensure transparent and sustainable post-implementation management, the project adopts a **community-academic partnership model** involving three main institutions, each with clear responsibilities and accountability mechanisms:

Stakeholder	Primary Responsibility	Secondary Support Role
Yayasan Gowa Raya	Acts as system custodian and manager of farm operations and hydroponic maintenance. Responsible for staff scheduling, solar power management, and record-keeping of revenue and expenditures.	Oversees training activities for new community members and ensures inclusion of women participants.
GR Farm (Hydroponic Partner)	Leads revenue collection, crop management, and produce sales . Maintains daily farm operations, ensures consistent vegetable quality, and tracks	Allocates 10 % of total annual income toward IoT maintenance and upgrades.

Stakeholder	Primary Responsibility	Secondary Support Role
	harvest output.	
Faculty of Informatics, Telkom University	Provides technical supervision, monitoring, and R&D support . Assists in troubleshooting, firmware upgrades, and periodic sensor calibration audits.	Hosts cloud backend, develops updated dashboards, and offers annual on-site inspection and capacity-building.

This structure clearly separates **financial management (GR Farm)** from **technical oversight (Telkom University)** and **community empowerment (Yayasan Gowa Raya)** — ensuring transparency, accountability, and operational continuity.

2. Revenue Collection and Reinvestment Mechanism

A **community-based reinvestment system** has been designed to sustain the hydroponic and solar infrastructure over time.

Revenue Sources:

- Sale of hydroponic vegetables to local markets, schools, and partner institutions.
- Sale of certified seedlings produced by community members to GR Farm.
- Reduced energy costs from solar utilization (USD 420 – 840 saved annually).

Revenue Management Workflow:

1. **Sales Transaction & Recording** – GR Farm handles all vegetable sales and issues receipts.
2. **Revenue Pooling** – All earnings are recorded in a dedicated *Project Maintenance Account* managed jointly by GR Farm and Yayasan Gowa Raya.
3. **Quarterly Allocation** –
 - 70 % for reinvestment in farm operations (nutrient solution, pump replacement, expansion).
 - 20 % distributed as income to community seedling partners and workers.
 - 10 % allocated to maintenance and technical upgrades (including sensor calibration and IoT components).

4. **Auditing & Transparency** – A financial report is produced quarterly by Yayasan Gowa Raya and verified by Telkom University to ensure accountability and IEEE-aligned transparency standards.

3. System Maintenance Responsibilities

System Component	Responsible Entity	Maintenance Schedule	Funding Source
pH, TDS, and Temperature Sensors	Trained technicians from Yayasan Gowa Raya	Calibration every 3 months; replacement every 12 months	10 % maintenance fund from revenue
Solar Panel & Battery System	GR Farm + Yayasan Gowa Raya	Visual inspection monthly; electrical test every 6 months	Reinvestment fund (70 %)
IoT Hardware (ESP32, relays, pumps)	Telkom University (supervision) + GR Farm	Preventive check every 3 months; repair if needed	Maintenance fund
Cloud Server & Mobile App	Faculty of Informatics, Telkom University	Continuous hosting and updates	University CSR / Research fund
Greenhouse Structure	Yayasan Gowa Raya + Community members	Cleaning and physical maintenance every 2 months	Operational budget

All maintenance activities will be logged through the **IoT dashboard's maintenance module**, allowing transparent tracking of replacement dates and calibration history.

4. System Upgrade and Innovation Plan

To ensure technological longevity and continual improvement, the project includes a **five-year upgrade roadmap** supported by academic collaboration and internal revenue allocation.

Year	Planned Upgrade	Expected Benefit
Year 1 (post-implementation)	Addition of “Vegetable Selector” feature to automate nutrient dosing based on crop type	Simplified operation for non-expert users

Year	Planned Upgrade	Expected Benefit
Year 2	Integration of weather data and predictive irrigation model	Further water and nutrient efficiency
Year 3	Cloud-based dashboard with AI-based yield forecasting	Enhanced productivity and market planning
Year 4	Integration of community e-commerce marketplace	Direct-to-consumer vegetable sales
Year 5	System replication to nearby villages under Yayasan Gowa Raya supervision	Expansion of community benefit and revenue base

Each upgrade phase will be supported by research projects from Telkom University's IoT Laboratory and senior design (capstone) programs, ensuring continuous innovation without additional external funding.

5. Summary of Post-Project Governance

Aspect	Responsible Institution	Sustainability Mechanism
Revenue Collection & Allocation	GR Farm + Yayasan Gowa Raya	70/20/10 revenue allocation model
Maintenance & Calibration	Yayasan Gowa Raya (trained technicians)	Routine schedule + maintenance fund
Technical Support & Upgrade	Faculty of Informatics, Telkom University	Yearly inspection + student R&D integration
Transparency & Reporting	Yayasan Gowa Raya	Quarterly financial & technical reports shared with IEEE Tech4Good

5. Gender Balance and Inclusion Strategy

1. Commitment to Gender Equality (SDG 5 Alignment)

The project team recognizes that achieving **gender balance** is essential to ensure equitable access to knowledge, income, and leadership opportunities within the community.

Aligned with **UN SDG 5: Gender Equality**, the project explicitly integrates gender inclusion in all stages — from planning and training to implementation and management.

The goal is not only to ensure fair participation but to empower women as **active technology users and decision-makers**, rather than passive beneficiaries.

2. Planned Gender Representation in Project Activities

Activity Area	Expected Gender Composition	Implementation Mechanism
IoT Device Training and System Maintenance	At least 30 % women participants from Yayasan Gowa Raya's school and pesantren network.	Priority recruitment of female teachers and students in STEM-related sessions.
Seedling Production and Greenhouse Operations	50 % female participation among local residents.	Women-led seedling groups trained in quality control, packaging, and digital sales via WhatsApp & mobile app.
Community Education and Public Workshops	Minimum 40 % women participants across all workshops.	Scheduling sessions compatible with domestic workloads and involving female facilitators.
Revenue and Financial Management	Shared roles between male and female representatives from Yayasan Gowa Raya.	A joint financial committee with at least one female treasurer ensures transparent revenue distribution.
Student Team Involvement	Gender balance target: 40–50 % female students among the 10 IEEE Student Members.	The Faculty of Informatics encourages female students to take lead roles in mobile app and UX design.

3. Practical Measures to Ensure Female Participation

To address potential barriers faced by women in rural and pesantren environments, the following concrete steps are included in the implementation plan:

1. Inclusive Scheduling:

Training sessions will be held at flexible hours (after 3 p.m. or weekends) to accommodate female participants with household duties.

2. Safe and Accessible Learning Spaces:

All training activities will take place within Yayasan Gowa Raya's educational compound — a familiar and safe environment for female participants.

3. **Women-Led Demonstration Units:**

Two hydroponic tables will be designated as “**Women’s Learning Units**”, managed by female participants to practice IoT monitoring, nutrient balancing, and system maintenance under direct mentorship.

4. **Mentorship from Female Academics:**

At least one female faculty member from Telkom University (from the Department of Computer Engineering or Informatics) will serve as **gender mentor and trainer**, encouraging women’s engagement in technical domains.

5. **STEM Career Exposure for Students:**

Female students from Yayasan Gowa Raya’s Junior and Senior High Schools will be introduced to IoT and renewable energy concepts through the “TryEngineering for Girls” module adapted from IEEE TryEngineering resources.

4. Gender-Disaggregated Impact Indicators

To monitor inclusion progress, the project will collect **gender-disaggregated data** and report it during the final evaluation:

Indicator	Baseline (2025)	Target (After Project)	Measurement Method
Female participants trained in IoT systems	0	≥ 12 women	Attendance records
Female-led seedling groups	0	≥ 3 groups	Community partner log
Female students engaged in STEM exposure	0	≥ 30 students	Workshop participation list
Female representation in decision-making roles	0	≥ 2 key roles (finance & training)	Project governance records
Perception of gender inclusivity (survey score)	—	≥ 80 % positive response	Post-project survey

5. Long-Term Women Empowerment Plan

Beyond the six-month implementation, gender inclusion will be sustained through:

- **Women's Hydroponic Cooperative:**
Formation of a women-led cooperative under Yayasan Gowa Raya to manage part of the greenhouse operations and future seedling distribution.
- **Micro-Entrepreneurship Training:**
Introduction of simple financial literacy and digital marketing workshops (WhatsApp Business, Canva for Agri-Products) to help women manage and sell their produce independently.
- **Integration into STEM Curriculum:**
Collaboration with Telkom University's Community Service Unit to maintain yearly workshops on IoT for Women in Agriculture, promoting continued skill development.
- **Monitoring and Evaluation:**
Telkom University will track the gender ratio annually and include the results in follow-up reports to IEEE Tech4Good and local partners.

6. Detailed Risk Analysis and Mitigation Plan

1. Overview

While the overall risk level for this project is considered *low to moderate*, the team has conducted a thorough assessment covering the **construction, operational, community, environmental, and reputational** dimensions.

Each identified risk includes specific **mitigation actions**, **responsible stakeholders**, and a **monitoring mechanism** to ensure that the system's performance, community safety, and IEEE's reputation remain protected throughout and beyond the project timeline.

2. Comprehensive Risk Table

Risk Category	Description of Risk	Potential Impact	Mitigation Strategy	Responsible Entity
A. Construction Risks	1. Greenhouse structure instability during heavy rainfall or wind, leading to potential damage.	Delay in installation, partial loss of equipment, or injury.	- Use galvanized steel frames and UV-resistant plastic with proper water drainage. - Employ professional constructors supervised by GR Farm and	Yayasan Gowa Raya + GR Farm

Risk Category	Description of Risk	Potential Impact	Mitigation Strategy	Responsible Entity
			Yayasan Gowa Raya. - Conduct pre-construction structural safety audit.	
	2. Solar panel installation hazard (electrical or fall risk).	Injury or damage to panels.	- Certified technicians for installation. - PPE use and IEEE-compliant safety checklist. - Grounding and waterproofing verification before activation.	Telkom Univ. + Certified Installer
B. Operational Risks	1. Sensor malfunction or calibration drift (pH, TDS).	Incorrect nutrient/pH adjustment → reduced yield.	- Schedule calibration every 3 months. - Maintain spare sensors (10 % redundancy). - Keep calibration log in IoT dashboard.	GR Farm + Telkom Univ.
	2. IoT connectivity interruption (power surge, Wi-Fi failure).	Data loss or delay in decision-making.	- Implement local SD card backup on ESP32. - Use solar power as backup for outages. - Offline mode on mobile app for critical alerts.	Telkom Univ. Student Team

Risk Category	Description of Risk	Potential Impact	Mitigation Strategy	Responsible Entity
	3. Pump or relay failure.	Nutrient flow disruption → crop loss.	- Provide 2 backup units. - Schedule preventive maintenance every 2 months. - Train operator to replace module manually.	GR Farm
C. Financial Risks	1. Delayed reinvestment of farm revenue for maintenance.	System degradation over time.	- Enforce 10 % maintenance fund policy (monitored quarterly). - Joint financial report by Yayasan Gowa Raya.	Yayasan Gowa Raya
	2. Fluctuation in vegetable market prices.	Reduced income and reinvestment capacity.	- Diversify crops (lettuce, kale, pak choi). - Engage in local contracts with restaurants and schools.	GR Farm
D. Community and Social Risks	1. Low community participation due to limited awareness or competing livelihoods.	Reduced local ownership and sustainability.	- Continuous engagement via training and community champions. - Incentivize participation through revenue-sharing and recognition certificates.	Yayasan Gowa Raya
	2. Gender imbalance	Missed SDG 5	- Reserved quotas	Project Lead

Risk Category	Description of Risk	Potential Impact	Mitigation Strategy	Responsible Entity
	in participation.	target and reduced inclusivity.	for women in all training sessions (min. 40 %). - Women-only learning units and flexible schedules.	+ Yayasan Gowa Raya
E. Environmental Risks	1. Water management failure (overflow or nutrient runoff).	Environmental contamination, reputational risk.	- Closed-loop hydroponic system with filtered discharge. - Periodic water testing for EC/pH before disposal.	GR Farm
	2. Solar battery disposal after lifespan.	Hazardous waste if unmanaged.	- Partner with certified battery recycling vendor. - Store used batteries safely until disposal.	Yayasan Gowa Raya
F. Reputational Risks (IEEE)	1. Miscommunication or misuse of IEEE logo/funds.	Institutional image risk.	- All publications, videos, and social media posts pre-approved by Telkom University's public relations unit and follow IEEE branding guidelines.	Project Lead
	2. Incomplete reporting or safety incident.	Non-compliance with IEEE Tech4Good standards.	- Appoint compliance liaison to ensure progress reports and safety documentation	Hilal H. Nuha (Project Lead)

Risk Category	Description of Risk	Potential Impact	Mitigation Strategy	Responsible Entity
			are submitted on schedule.	

3. Environmental, Health, and Safety (EHS) Compliance

- **Safety Training:** All community and student participants will undergo basic EHS orientation (equipment handling, electrical safety, and PPE).
- **Electrical Compliance:** All electrical installations will follow Indonesian SNI 04-0225-2000 standards, equivalent to IEC 60364, ensuring proper grounding and surge protection.
- **Fire and Emergency Readiness:** Fire extinguisher and emergency signage will be installed in the greenhouse.
- **Environmental Compliance:** The hydroponic system uses a **closed nutrient loop**, ensuring zero wastewater contamination to nearby soil or rivers.

4. Monitoring and Reporting Mechanism

Stage	Risk Monitoring Activity	Frequency	Reporting Entity
Pre-construction	Site and structural safety inspection	Once before construction	GR Farm + Telkom Univ.
Implementation	IoT system and solar installation audit	Weekly	Project Team
Operation	Maintenance and calibration logging	Monthly	Yayasan Gowa Raya
Financial Oversight	Revenue and expenditure review	Quarterly	GR Farm + Faculty of Informatics
Final Evaluation	Comprehensive risk reassessment	End of 6th month	IEEE Tech4Good Report Lead

All incidents, near-misses, or failures will be logged into the **Project Risk Register**, hosted on Telkom University's project management system and accessible to IEEE upon request.

5. Contingency and Adaptation Plan

Should any major disruption occur (e.g., natural disaster, equipment damage, or social unrest), the following contingency framework will be activated:

1. **Damage Assessment Team** (GR Farm + Telkom Univ.) mobilized within 48 hours.
2. **Emergency fund (10 % of budget reserve)** used for immediate repairs.
3. **Community notification protocol** through Yayasan Gowa Raya's communication channel.
4. **IEEE update report** submitted within 14 days detailing corrective actions and recovery progress.

This ensures rapid response, transparency, and continuity of project operations.

6. Summary of Risk Management Framework

Dimension	Status	Key Safeguard
Construction & Equipment	Moderate risk	Structural safety audit & certified installers
Operation & Maintenance	Low risk	Preventive schedule, redundancy design
Financial & Community	Moderate risk	Transparent fund governance & incentive system
Environmental	Low risk	Closed-loop irrigation & safe battery disposal
Reputational	Very low risk	IEEE compliance and periodic reporting

Conclusion

The project team has addressed all reviewer clarifications with detailed governance, measurable impact indicators, sustainability models, and community empowerment strategies. These enhancements ensure technical reliability, financial independence, and strong gender-inclusive participation for long-term sustainability aligned with IEEE Tech4Good objectives.

Letter of Support

1. LoS from GR Farm



GR FARM Urban Hydro Farm

Jl. Sirajuddin Rani No.39 A, Sungguminasa, Kabupaten Gowa.
Email : grcorp.farm@gmail.com | Telp. 081235353677

LETTER OF SUPPORT

Hereby,

I, Irfan Fadly, the partner of GR Farm, located at Yayasan Gowa Raya, am pleased to express our full support for the collaboration with Telkom University Team (Bandung Campus) on the IEEE Communication for All and Tech4Good 2025 project titled "IoT-based Hydroponic Farming and Solar Energy Integration for Sustainable Agriculture."

Our farm currently operates eight hydroponic tables, each of which yields approximately 30 kg per harvest. The harvest cycle is completed every seven weeks, with produce being sold for IDR 35,000 to IDR 45,000 per kilogram. The success of this venture has been significantly enhanced by our ongoing partnership with "Makan Bergizi Gratis" in Mamuju, West Sulawesi, contributing to local food security efforts and improving access to fresh, nutritious food for the community.

Through this collaboration, we aim to expand our capacity by adding an additional 20 hydroponic tables, which will increase our production output and further contribute to strengthening food resilience in the region. We are confident that the implementation of IoT technology in our farming practices will help optimize the monitoring and management of our hydroponic system, enhancing its efficiency and sustainability. We fully support the objectives of this project and the team's efforts in enhancing food security, promoting sustainability, and improving agricultural practices in the local community. We look forward to continued collaboration and are excited about the potential impact of this initiative.

Thank you for your commitment to this project, and we are eager to see the positive changes it will bring to our community.

Bonto-Bontoa, 10 September 2025

Lead of GR Farm


Irfan Fadly
GR farm
#enjoy ur fresh food
@grfarm - 0812-3535-3677 @grfarm

2. LoS from Yayasan Gowa Raya



YAYASAN PENDIDIKAN GOWA RAYA

Membina :

1. STIK KHARISMA GOWA RAYA (S1 Keperawatan,D3 Kebidanan,D3 Farmasi)
2. AKPER KHARISMA GOWA RAYA
3. SMK KEPERAWATAN KHARISMA GOWA RAYA
4. SMP-SMA GOWA RAYA
5. SMK TEKNOLOGI GOWA RAYA

Alamat Jl.Nuri No.9 Kel. Bonto Bontoa Sunggiminasa, Kabupaten Gowa

LETTER OF SUPPORT

Hereby,

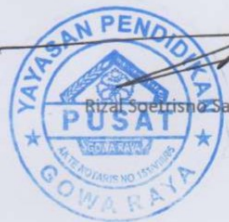
I, Rizal Soetrismo Saillellah, act as the leader of Yayasan Gowa Raya, Bonto-Bontoa Village, Gowa Regency, South Sulawesi, expressed my full support for the project proposed by the Telkom University (Bandung Campus) team that proposed the IEEE Communication for All and Tech4Good 2025 program with the topic "IoT-based Hydroponic Farming and Solar Energy Integration for Sustainable Agriculture"

The project aims to improve food security and self-reliance with GR Farm in Yayasan Gowa Raya located in Bonto-Bontoa Village. Through the use of vacant land and environmentally friendly technology, integrating hydroponic systems and solar energy.

We consider that this initiative is highly significant and offers numerous benefits, not only in meeting sustainable food needs but also in providing entrepreneurship and agribusiness skills training to the student of Yayasan Gowa Raya. This system is estimated to produce 120 kg of vegetables every two weeks, with the potential to open new revenue opportunities through the sale of the crops. Thus, Yayasan Gowa Raya fully supports the Telkom University team in implementing this project as a serious effort to contribute to food security, energy independence, and sustainable development for communities in need.

Bonto-Bontoa, 10 September 2025

Lead of Yayasan Gowa Raya



Rizal Soetrismo Saillellah

3. LoS from Faculty of Informatics Telkom University



LETTER OF SUPPORT

Hereby,

I am Dr. Kemas Muslim Lhaksmana, Dean of the Faculty of Informatics, Telkom University (Bandung Campus), fully support for our team from Telkom University that submits to the IEEE Tech4Good 2025 Program entitled: *"IoT-based Hydroponic Farming and Solar Energy Integration for Sustainable Agriculture."*

Yayasan Gowa Raya is a private foundation that consists of a Junior High School (SMP), Senior High School (SMA), and a pesantren (Islamic boarding school) for underprivileged communities in Bonto-Bontoa Village. Therefore, we collaborated to enhance the food resilience within this organization by applying eco-friendly farmland.

This initiative is a strategic and impactful program that aligns with the Faculty of Informatics' mission and Telkom University's vision for supporting Sustainable Development Goals (SDGs) and community empowerment. The proposed activities are the integration of IoT-based hydroponic farming systems and solar energy solutions. These activities are highly relevant in addressing food resilience and environmental sustainability.

As a form of institutional commitment, the Faculty of Informatics, Telkom University, is committed to providing support in the implementation of this project, particularly in the form of technical expertise and administrative assistance required for project execution and reporting.

We believe that the outcomes of this program will not only contribute to the welfare of Yayasan Gowa Raya and its community but also exemplify an effective model of university-community collaboration in delivering innovation-driven solutions for real-world challenges.

Bandung, 12 September 2025

Dean of the Faculty of Informatics, Telkom University

A blue ink signature of Dr. Kemas Muslim Lhaksmana, written over the Telkom University logo.

Dr. Kemas Muslim Lhaksmana

4. LoS from Telkom University



LETTER OF SUPPORT

Hereby,

I am Prof. Suyanto, the Rector of Telkom University, fully support for our team from Telkom University (Bandung Campus) lead by Hilal Hudan Nuha, S.T., M.T., Ph.D. that submits to the IEEE Tech4Good 2025 Program entitled: "IoT-based Hydroponic Farming and Solar Energy Integration for Sustainable Agriculture". Yayasan Gowa Raya is a private foundation that consists of a Junior High School (SMP), Senior High School (SMA), and a pesantren (Islamic boarding school) for underprivileged communities in Bonto-Bontoa Village. Therefore, we collaborated to enhance the food resilience within this organization by applying eco-friendly farmland.

This initiative is a strategic and impactful program that aligns with the university's mission for supporting Sustainable Development Goals (SDGs) and community empowerment. The proposed activities are the integration of IoT-based hydroponic farming systems and solar energy solutions. These activities are highly relevant in addressing food resilience and environmental sustainability.

As a form of institutional commitment, Telkom University (Bandung Campus) is committed to providing support in the implementation of this project, particularly in the form of funding assistance for technical and administrative personnel required for project execution and reporting.

We believe that the outcomes of this program will not only contribute to the welfare of Yayasan Gowa Raya and its community but also exemplify an effective model of university-community collaboration in delivering innovation-driven solutions for real-world challenges.

Bandung, 12 September 2025

Rector of Telkom University

Prof. Dr. Suyanto, S.T., M.Sc.

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